

COURSE SYLLABUS

CS486 – INTRODUCTION TO DATABASE SYSTEMS

1. GENERAL INFORMATION

Course name:	Introduction to Database Systems
Course name (in Vietnamese):	Nhập môn các hệ cơ sở dữ liệu
Course ID:	CS486
Knowledge block:	Compulsory
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	N/A
Prior course:	Discrete Mathematics, Data Structure
Instructors:	Dr. Le Thi Nhan

2. COURSE DESCRIPTION

The course is designed to provide students the overview of the needs for databases in enterprises, as well as other organizations. The course will provide the background knowledge of database systems on where the relational model is emphasized. Moreover, the techniques, tools and skills are provided for students to design, manipulate, and exploit the database via a relational database management system. The course also mentions future trends in database systems research.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Explain roles and basic concepts of a database in an information system	LO1
G2	Apply common data models for modeling data	LO2
G3	Master query languages to exploit the relational database	LO2
G4	Design a good database schema	LO4
G5	Describe how main components of Database Management System (DBMS) work	LO1
G6	Practice skills of building databases and queries in DBMS	LO6

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Interpret basic concepts of database systems	I
G2.1	Apply the Entity Relationship Model for data modeling in the conceptual level	T
G2.2	Apply the Relational Data Model for data modeling in the logical level	T
G3.1	Use Relational Algebra to manipulate the data	TU
G3.2	Use Relational Calculus to query the data	TU
G3.3	Use Structured Query Language (SQL) to define and manipulate the data	T
G4.1	Describe integrity constraints of the database	T
G4.2	Interpret the quality of a database schema	T

G5.1	Explain how a transaction manager works	T
G5.2	Explain how schedulers and locking schemas work	T
G5.3	Explain how data recovery methods work	T
G6.1	Utilize a Database Management System (DBMS) to define database schemas and exploit data	T

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessment
1	Overview database system	G1	Lecturing, Q&A, (Chapter 1) [2]	
2	Entity relationship model	G2	Lecturing, Discussion (Chapter 3) [2]	HW1
3	Relational data model	G2	Lecturing, Discussion (Chapter 2) [1]	HW2
4	Query languages	G3, G6	Discussion (Chapter 2, 5, 6, 8) [1] (Chapter 6) [2]	HW3
5	Query languages (cont.)	G3, G6	Discussion (Chapter 2, 5, 6, 8) [1] (Chapter 6) [2]	HW3
6	Integrity constraint	G4, G6	Lecturing, Discussion (Chapter 7) [1]	HW4
7	Functional dependency and normal forms	G4	Lecturing, Discussion	HW4

			(Chapter 3) [1]	
8	Transaction, scheduler and locking	G5, G6	Lecturing, Discussion (Chapter 17) [1]	HW5
9	Recovery methods	G5	Lecturing, Discussion (Chapter 18) [1]	HW5
10	Other database issues	G1	Lecturing, Q&A, Discussion	

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities	Assessment
1	Setup	G6	Demonstration	LW1
2	Basic SQL operations	G3	Demonstration	LW2
3	SQL data definition	G3	Demonstration	LW3
4	Complex SQL queries	G3	Demonstration	LW4
5	Stored procedure and Trigger	G3, G4	Demonstration	LW5
6	Midterm exam	G3, G6	Coding	
7	Transaction and Isolation level	G5	Demonstration	LW6
8	Extras	G3, G4, G5	Demonstration	LW7
9	Database application	G6	Demonstration	LW8
10	Final exam	G3-G6		

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A11	Exercise	Small exercises in class for each topic	G1-G5	10%
A12	Homework: HW1~HW5	Reading comprehension and writing reports in English Practicing based on knowledge taught in class	G1-G5	10%
A2	Lab			40%
A21	Weekly lab work: LW1~LW8	Practicing based on knowledge taught in class Doing small quizzes to consolidate the knowledge	G3.3, G4, G5, G6	16%
A22	Lab midterm exam	In-class programming exam on computer Closed book exam	G3.3	12%
A23	Lab final exam	In-class programming exam on computer Closed book exam	G3.3, G4, G5	12%
A3	Exams			40%
A31	Finals exam	Describe the understanding of different topics, analyze to solve problems Closed book exam	G1-G5	40%

7. RESOURCES

Textbooks

- [1]. Database Systems: The Complete Book (2nd Edition), Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, Prentice Hall, 2008.
- [2]. Fundamentals of Database Systems (7th Edition), Ramez Elmasri, Shamkant B. Navathe, Addison Wesley, 2017.

Others

- [3]. Database system concepts (7th Edition), Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill, 2020.
- [4]. Database Management System (3rd Edition), Raghu Ramakrishnan and Johannes Gehrke, McGraw-Hill, 2003.

Software

- [1]. MS SQL Server: T-SQL
- [2]. Visual studio.NET: C#, ASP.NET Framework

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.